Gabriel Cirker

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EDUCATION

University of Michigan, *Ann Arbor*.....Expected May 2028

- **Double major:** Physics and Mathematics
- **Current courses:** MATH 463: Mathematical Biology; PHYSICS 411: Computational Physics

St. John's University, *Queens, NY*.....2025 - 2026
Prior undergraduate study

Physics and Mathematics; Data Science minor during prior study

GPA 4.00 through Fall 2025; Dean's List, Fall 2025

Earlier coursework: Calculus II, Calculus III, and University Physics I.

Spring 2026 courses: Linear Algebra, Ordinary Differential Equations, Optics, University Physics II, and Concepts in Mathematics with Applications.

University of San Diego, *San Diego, CA*.....Aug 2023 - Dec 2024
Prior undergraduate study

Biomedical Engineering studies; recruited member of the men's rowing team.

RESEARCH EXPERIENCE

HIV mRNA Secondary Structure and Energy Modeling
St. John's University.....Jan - Jun 2026
Biostatistical Researcher

- Analyzed HIV RNA samples in R to investigate associations between sequence and secondary-structure features and predicted minimum free energy.
- Examined sequence length, paired-region and block lengths, nucleotide composition, and purine-rich motif placement, including GGAG.
- Built structure-derived predictors and compared regression models with diagnostic visualizations; summarized patterns in a project overview.

SELECTED COMPUTATIONAL PROJECTS

Modeling Immune Activation and Self-Tolerance
Three-person course project, MATH 463, University of Michigan.....Fall 2026
Coauthors: Gabriel Cirker, Kennedy Jamison, and Cecilia Pirozzi; instructor: Naoki Masuda.

- Coauthored a report adapting a published immune-activation model to three dynamic states: conventional T cells, regulatory T cells, and IL-2.
- The group used SciPy Radau integration, a tighter-tolerance numerical check, and parameter sweeps to examine the effect of regulatory-cell input on simulated dynamics.
- The study compared its baseline with a digitized published curve, distinguishing image-resolution agreement from access to the original numerical data.
- The group applied sparse identification of nonlinear dynamics (SINDy) with a quadratic feature library, separate validation for threshold selection, and a held-out initial condition for testing.
- The group recovered all 12 true terms in the clean simulated baseline. A quieter regime omitted one term despite accurate trajectories, distinguishing prediction from equation recovery.

Project, report, and supporting material

SELECTED COMPUTATIONAL PROJECTS CONTINUED

Computational Biomechanics of Rowing
Independent project | Project, report, and archive

- Developed a physics-based simulation of mechanical energy transfer from rowing strokes to boat motion, with explicit treatment of losses and efficiency.
- Used Python and NumPy/SciPy to explore drag, stroke rate, and force-time assumptions; prepared analysis, figures, and a technical report.

Research focus: mechanical modeling of rowing technique and performance.

Mathematical Optimization of Lightweight Bridge Structures
Independent project | Project, report, and archive

- Compared equal-mass solid, hollow, and tapered beam geometries using Euler-Bernoulli modeling and numerical calculations.
- Investigated static deflection and buckling through linear-system and generalized-eigenvalue formulations.

Documented modeling assumptions, numerical results, and design comparisons in a technical report.

GPU Compute Rig Optimization
Independent hardware project

- Designed, assembled, and tuned an 8-GPU compute system.
- Used power and thermal monitoring to assess performance, efficiency, and configuration changes.

TEACHING AND MENTORING

Assistant Rowing Coach
Friends of Port Rowing..... Jun 2023 - Aug 2025

- Coached novice through experienced athletes in technique, biomechanics, and race strategy.
- Supported training planning, athlete development, and safe practice.

Rowing Leadership and Service
Friends of Port Rowing..... 2019 - 2022

- Served as varsity team captain and helped lead training, team logistics, and athlete mentoring.
- Provided adaptive rowing instruction through the Inclusion Rowing Program.

SELECTED HONORS

Dean's List, St. John's University, Fall 2025.

First place, Freshman Engineering Bridge Competition, University of San Diego, May 2024.

USRowing Summer National Champion, U17 quad, 2021.

Youth Nationals silver medalist, U17 quad, 2021.

TECHNICAL SKILLS

Programming and analysis: Python, R; NumPy, SciPy, pandas, Matplotlib, tidyverse, ggplot2.

Methods: numerical integration, linear systems, eigenvalue problems, regression, parameter sweeps, sparse system identification, and scientific visualization.

Tools: Git, Jupyter, LaTeX, Excel.